Dealing With NA's - Lab 8			
function	**description**	**example**	
clean_names()	manage inconsistencies and weird characters in column names		
miss_var_summary	provides a summary of NA's across data frame		
na_if()	replacing X with X	mutate(max_life=na_if(max_life, 0)	replace 0's with NA's
replace_with_na	allows you to replace the known NA valuesm(e.g. -999, 0, etc) with "NA"	replace_with_na(reaplce = list(newborn = "not measured, weaning" = -999, etc....	
replace_with_na_all(condition = ~.x == NA value)	use naniar, to replace a specific value with NA across the entire data set		
na = c("NA", " ", ".", "-999", "not measured)	deaing with NA's in advance, use once you are sure how NA's are treated	read_csv("path", na = c(.......))	
na.omit()	removes Nas	this will drop NA's when using pivot longer	

Tidy Data and Pivot_XX() - Lab 8/9			
function	**description**	**example**	
tidy data = each variable has it own column, each obs has its own row, and each val has its own cell			
pivot_longer(names_to = , values_to =)	makes a new column called 'X' and valyes moved to a new column called 'Y'		
pivot_longer(-c(.. , .., ..), names_to = , values_to =)	specific columns to stay fixed while using pivot_longer		
pivot_longer(names_to = c("X","Y") , names_sep "by whatever the column name is sep by", values_to =)	names_sep helps pull apart two observations in a column name		
as_tibble()	this transforms the matrix into a data frame	separate(patient, into= c("patient", "sex", sep="_"))	
separate()	seperates two values per cell into its own column		
unite()	combines to column names into one	unite(patient_sex, "patient", "sex", sep="_"	
pivot_wider(names_from= , values_from =)	used when you have an observation scattered across multiple rows		
names_from	the observations under key will become new column		
values_from	the values under value will be moved to the new column		

Data Visualization - ggplot and aesthetics- Lab 9/10/11			
Type of Data	**Types of ggplots to make**		
Discrete - quantitative data that only contains integers	**geom_bar , geom_col, geom_boxplot**		
Continuous - quantitative data that can take any numerical value	**geom_histogram, geom_density, geom_point**		
Categorical - qualitative data that can take on a limited number of values	**geom_bar, geom_boxplot, facet_wrap, facet_grid**		
function within ggplot	**description**		
geom_point()	makes a scatter plot		
geom_jitter()	similar to geom_point but helps with over plotting by adding random noise to the data		
geom_smooth()	add a regresssion (best of fit) line; add under geom_point	geom_point() + geom_smooth(method=lm, se=T)	
geom_bar()	simplest type of bar plot cunts the number of observations in categorical variable	homerange % ggplot(aes(x=trophic.guild))+ geom_bar()	
geom_col()	allows us to specify an x-axis and y-axis		
coord_flip()	flips axes of the plot		
geom_boxplot()	visualize range of values, x= categorica, y =is the range		
scale_y_log10()	changes the y axis scale		
reorder()	reordering x axis variables	ggplot(aes(x=reorder(taxon,mea_mean_mass), y=mean_mass))	order x values by y values
geom_point(size= #)	controls the size of the point		
geom_point(aes(color=variable), size=1.5)	color the points by a categorical variable		
geom_point(aes(shape=variable, color=variable),size=1.50)	map shapes to another categorical variable		
ggplot(aes(x=taxon, fill = taxon))	filling by a different categorical variable, and also builds a stacked bar plor		
ggplot(aes(x=taxon, fill = taxon))+ geom_bar(position = "dodge")	unstacks the bargraphs and has the counts side by side		
group()	make individual box plots for each variable	ggplot(aes(x=class, y=log10.mass, group = taxon)+ geom_boxplot	
mutate(colname=as.factor(colname))			
geom_histogram(bins= #)	controls the number of bins in histogram		
geom_density + geom_histogram	creates an overlaying density plot over histogram	geom_histogram(aes(after_stat(density)), fill = "deepskyblue", alpha=0.4, color ="black")+ geom_density(color = "red")	
quantcut()	finds the quartiles of a variable	quantcut(homerange$log10.mass) table(quartiles)	
case_when	turns a contnuous thing into categorical	mutate(mass_category = case_when(log10.mass = 1.7 ~ "small" log10.mass > 1.7 & log 10.mass <= 3.3 ~ "medium" log10. mass > 3.3 ~ "large")	
facet_wrap()	ribbon of panels by a specified categorical variable	facet_wrap(~trophic.guild, ncol =2, labeller=label_both)	
facet_grid()	it can be arranged in rows or columns (rows, ~ columns.)	facet_grid(trophic.guild~.)	comparison of two categorical variables (facet_grid(trophic.guild~thermoregulation)

Aesthetics - Lab 11/12/13			
function	**description**		
labs(title = , x= , y=)	gives labels to the graph	labs(title = "Elephant Age vs Height", x= "Age", y="Height"	
theme(plot.title=element_text(size=rel(1.5), hjust=.5))	increases the relative size of the title and centers the title in middle/half way		
theme(axis.text.x = element_text(angle = 60, hjust=1))	adjusts the angle of the x axis variable names		
color =	controls the the outline color of a box		
fill =	controls the color inside the box		
alpha =	controls the transparency		
theme(legend.position = "top/bottom", axis.text.x = element_text(angle=60, hjust=1)	manipulates legends		
ls("package:ggthemes")[grepl("theme_", ls("package:ggthemes"))]	lists the ggthemes	theme_stata()+ theme(legend.position).......etc	need to install.packages("ggthemes"), library(ggthemes)

Joins - Lab 13			
join_type(first table, second table, by = columnTojoinOn)			
function	**description**		
inner_join()	what is shared between them		
left_join()	includes first table and what is shared between them		
right_join()	incudes second table, and what is shared between them		
full_join()	includes everything from first and second table, including what is shared		
anti_join()	only includes one table; undo		